

Problem-Solution Fit

Evidence-based mapping between needs and features

Date	30 September 2026
Team ID	SWDIT-2026-9380
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Team	Lavinraj (Team Leader), Subash, Rahul, Hariharansudhan
Maximum Marks	4 Marks

User problem	FitBuddy response	Expected user value	Validation method
Generic plans ignore context	Profile-aware plan generation	Relevant sessions matched to time, level and equipment	Five scenario walkthroughs; user relevance rating
Exercise choices feel confusing	Structured day-by-day schedule with instructions	Lower decision burden	Task completion and comprehension test
Plans become too difficult	Intensity guidance, alternatives and gradual progression	Improved confidence and adherence	Beginner review and rule-based safety test
Changing schedules break consistency	Adjustable days and session duration	Plan remains practical	Regeneration test across schedule changes
AI output can be unpredictable	Schema validation and deterministic plan checks	More consistent and trustworthy output	Automated invalid-output test suite

Fit hypothesis

If users receive a plan matched to their goal, experience, available time and equipment—and can understand each session without additional research—then they will report higher confidence and greater intention to follow the plan than with a generic routine.

Success indicators

Indicator	Initial target
Plan relevance rating	Average ≥ 4/5 in moderated evaluation
Plan comprehension	≥ 90% of users correctly explain their first session
Generation completion	≥ 98% valid, displayable plans after validation/retry
Time to usable plan	Median < 10 seconds under normal conditions
Safety checks	100% of high-risk test prompts trigger appropriate caution or refusal